

MECH5200 Numerical Methods for PDEs : *Course Introduction*

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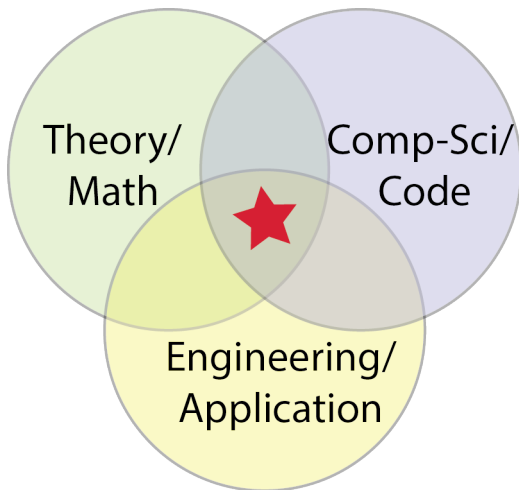
Introductions

- **MECH5200 Numerical Methods for PDEs** : Graduate & Seniors
- **Live Session**: Wednesday: 6:30PM - 9:20PM – DAN-220
- **Email**: Valentin_Boutrouche@uml.edu

What is the purpose of this course?

- Specifically, you will learn:
 - **[FDM]** Finite Difference Methods: Theory, Discretization, and Numerical Solutions.
 - **[FVM]** Finite Volume Methods: Theory, Discretization, and Numerical Solutions.
 - **[FEM]** Finite Element Methods: Theory, Discretization, and Numerical Solutions.
 - **[BEM]** Boundary Element Methods: Theory, Discretization, and Numerical Solutions.
- And supporting this will be:
 - **[MAT]** Numerical methods for solving linear and nonlinear systems.
 - **[GEO]** Discrete Geometry Representation and Mesh Generation

This course is a combination of:



Blended and Flipped Classroom

- Semi-blended classroom:
 - Two to three $\times 10 - 20$ minute videos a week on core concepts
 - Watch before class
 - Meet in class
 - Blended instruction and coding (MATLAB).
 - In-class code/math activities
 - Questions and explorations
 - High level of participation expected
 - Identify weaknesses/learning/growth opportunities
 - Challenge our assumptions and fears – supportive environment
 - Go home and do the homework.
- Thoughts/concerns?

Grades and Deliverables

- 60% **Homework** – individually performed homework problems with computer programming (4-6 total "mini-project" assignments), submitted to **GradeScope**.
- 20% **Computer Programming Project** – A group/individual project in a field of your choice. 2-4 students in one group.
- 20% **In-class exercises** – in-class exercises and assignments.

Grades and Deliverables

- Homework will cover main concepts discussed in online videos and in class (theory, coding and application).
 - All homeworks & projects will require MATLAB or another programming language (MATLAB/C++/Fortran etc.)
 - Homework will be due on GradeScope (through Bb).
 - Homework policy: You **may not** code/implement homework problems together outside of class.
 - Computer code that is similar to other students' → grade of zero.
 - Study and workgroups are permitted; however, code and written deliverable to hand-in **must** be *individual*.
 - Any discussions/references used for HW content must be clearly referenced in detail.

Since the homework is the source of the majority of your grade, a zero-tolerance policy will be in effect in accordance with the UML academic integrity policy:

<http://www.uml.edu/Catalog/Undergraduate/Policies/Academic-Integrity.aspx>

Frequently Asked Questions

Questions?

- *Do I need to type my answers for questions that are not implemented on a computer?*
- *Do I need to type my project proposal and final report?*
- *How do I hand in the portions of the homework that are implemented on a computer?*
- *Can I ask for additional computer coding help?*
- *Do I need to know MATLAB?*
- *What if I know no programming languages?*

Online Materials and Textbook(s)

- MIT Opencourseware : **16.920**: These notes have additional information you may find useful, 16.910, 16.901
 - <http://ocw.mit.edu/>
- Reference books are on syllabus and will be listed in Bb.
- Lecture recordings and pre-lecture mini-lecture videos will be posted on Echo360 and links shared via Blackboard site.

Approximate Schedule

- Introduction, MATLAB & some linear algebra – (1-2 weeks)
- Finite difference methods for elliptical, parabolic and hyperbolic equations (3-4 weeks)
- Finite volume methods for hyperbolic equations/conservation laws (3 weeks)
- Finite element methods (3-4 weeks)
- Boundary element methods/DG (1-2 weeks)

Questions?

Any questions...?